

1 The Limiting MPC's

For $m_t > 0$ we can define $e_t(m_t) = c_t(m_t)/m_t$ and $a_t(m_t) = m_t - c_t(m_t)$ and the Euler equation (4) can be rewritten

$$\begin{aligned} e_t(m_t)^{-\gamma} &= \beta \mathbb{R} \mathbb{E}_t \left[\left(e_{t+1}(m_{t+1}) \left(\frac{\overbrace{\mathcal{R}a_t(m_t) + \mathcal{G}_{t+1}\xi_{t+1}}^{=m_{t+1}\mathcal{G}_{t+1}}}{m_t} \right) \right)^{-\gamma} \right] \\ &= (1 - \wp) \beta \mathcal{R} m_t^\gamma \mathbb{E}_t \left[(e_{t+1}(m_{t+1}) m_{t+1} \mathcal{G}_{t+1})^{-\gamma} \mid \xi_{t+1} > 0 \right] \\ &\quad + \wp \beta \mathcal{R}^{1-\gamma} \mathbb{E}_t \left[\left(e_{t+1}(\mathcal{R}_{t+1} a_t(m_t)) \frac{m_t - c_t(m_t)}{m_t} \right)^{-\gamma} \mid \xi_{t+1} = 0 \right]. \end{aligned}$$

Consider the first conditional expectation in (4), recalling that if $\xi_{t+1} > 0$ then $\xi_{t+1} \equiv \theta_{t+1}/(1 - \wp)$. Since $\lim_{m \downarrow 0} a_t(m) = 0$, $\mathbb{E}_t[(e_{t+1}(m_{t+1}) m_{t+1} \mathcal{G}_{t+1})^{-\gamma} \mid \xi_{t+1} > 0]$ is contained within bounds defined by $(e_{t+1}(\underline{\theta}/(1 - \wp)) \mathcal{G} \underline{\psi} \underline{\theta}/(1 - \wp))^{-\gamma}$ and $(e_{t+1}(\bar{\theta}/(1 - \wp)) \mathcal{G} \bar{\psi} \bar{\theta}/(1 - \wp))^{-\gamma}$ both of which are finite numbers, implying that the whole term multiplied by $(1 - \wp)$ goes to zero as m_t^γ goes to zero. As $m_t \downarrow 0$ the expectation in the other term goes to $\bar{\kappa}_{t+1}^{-\gamma} (1 - \bar{\kappa}_t)^{-\gamma}$. (This follows from the strict concavity and differentiability of the consumption function.) It follows that the limiting $\bar{\kappa}_t$ satisfies $\bar{\kappa}_t^{-\gamma} = \beta \wp \mathcal{R}^{1-\gamma} \bar{\kappa}_{t+1}^{-\gamma} (1 - \bar{\kappa}_t)^{-\gamma}$. Exponentiating by γ , we can conclude that

$$\begin{aligned} \bar{\kappa}_t &= \wp^{-1/\gamma} (\beta \mathcal{R})^{-1/\gamma} \mathcal{R} (1 - \bar{\kappa}_t) \bar{\kappa}_{t+1} \\ \underbrace{\wp^{1/\gamma} \mathcal{R}^{-1} (\beta \mathcal{R})^{1/\gamma}}_{\equiv \wp^{1/\gamma} \mathbf{P}/\mathcal{R}} \bar{\kappa}_t &= (1 - \bar{\kappa}_t) \bar{\kappa}_{t+1} \end{aligned}$$

which yields a useful recursive formula for the maximal marginal propensity to consume:

$$\begin{aligned} (\wp^{1/\gamma} \mathbf{P}/\mathcal{R} \bar{\kappa}_t)^{-1} &= (1 - \bar{\kappa}_t)^{-1} \bar{\kappa}_{t+1}^{-1} \\ \bar{\kappa}_t^{-1} (1 - \bar{\kappa}_t) &= \wp^{1/\gamma} \mathbf{P}/\mathcal{R} \bar{\kappa}_{t+1}^{-1} \\ \bar{\kappa}_t^{-1} &= 1 + \wp^{1/\gamma} \mathbf{P}/\mathcal{R} \bar{\kappa}_{t+1}^{-1}. \end{aligned}$$

As noted in the main text, we need the WRIC L.4 for this to be a convergent sequence:

$$0 \leq \wp^{1/\gamma} \mathbf{P}/\mathcal{R} < 1, \quad (1)$$

Since $\bar{\kappa}_T = 1$, iterating (1) backward to infinity (because we are interested in the limiting consumption function) we obtain:

$$\lim_{n \rightarrow \infty} \bar{\kappa}_{T-n} = \bar{\kappa} \equiv 1 - \wp^{1/\gamma} \mathbf{P}/\mathcal{R} \quad (2)$$

and we will therefore call $\bar{\kappa}$ the ‘limiting maximal MPC.’

The minimal MPC’s are obtained by considering the case where $m_t \uparrow \infty$. If the FHWC holds, then as $m_t \uparrow \infty$ the proportion of current and future consumption that will be financed out of capital approaches 1. Thus, the terms involving ξ_{t+1} in (1) can be neglected, leading to a revised limiting Euler equation

$$(m_t e_t(m_t))^{-\gamma} = \beta R \mathbb{E}_t \left[\left(e_{t+1}(a_t(m_t) \tilde{\mathcal{R}}_{t+1}) (Ra_t(m_t)) \right)^{-\gamma} \right]$$

and using L’Hôpital’s rule $\lim_{m_t \rightarrow \infty} e_t(m_t) = \underline{\kappa}_t$, and $\lim_{m_t \rightarrow \infty} e_{t+1}(a_t(m_t) \tilde{\mathcal{R}}_{t+1}) = \underline{\kappa}_{t+1}$ so a further limit of the Euler equation is

$$\begin{aligned} (m_t \underline{\kappa}_t)^{-\gamma} &= \beta R (\underline{\kappa}_{t+1} R (1 - \underline{\kappa}_t) m_t)^{-\gamma} \\ \underbrace{R^{-1} \mathbf{P}}_{\equiv \mathbf{P}/R=(1-\underline{\kappa})} \underline{\kappa}_t &= (1 - \underline{\kappa}_t) \underline{\kappa}_{t+1} \end{aligned}$$

and the same sequence of derivations used above yields the conclusion that if the RIC $0 \leq \mathbf{P}/R < 1$ holds, then a recursive formula for the minimal marginal propensity to consume is given by

$$\underline{\kappa}_t^{-1} = 1 + \underline{\kappa}_{t+1}^{-1} \mathbf{P}/R \quad (3)$$

so that $(\{\underline{\kappa}_{T-n}^{-1}\})_{n=0}^{\infty}$ is also an increasing convergent sequence, and we define

$$\underline{\kappa}^{-1} \equiv \lim_{n \uparrow \infty} \underline{\kappa}_{T-n}^{-1} \quad (4)$$

as the limiting (inverse) marginal MPC. If the RIC does *not* hold, then $\lim_{n \rightarrow \infty} \underline{\kappa}_{T-n}^{-1} = \infty$ and so the limiting MPC is $\underline{\kappa} = 0$.

For the purpose of constructing the limiting perfect foresight consumption function, it is useful further to note that the PDV of consumption is given by

$$c_t \underbrace{\left(1 + \mathbf{P}/R + \mathbf{P}/R^2 + \dots \right)}_{=1+\mathbf{P}/R(1+\mathbf{P}/R\underline{\kappa}_{t+2}^{-1})\dots} = c_t \underline{\kappa}_{T-n}^{-1}.$$

which, combined with the intertemporal budget constraint, yields the usual formula for the perfect foresight consumption function:

$$c_t = (b_t + h_t) \underline{\kappa}_t \quad (5)$$